

CSE331 - Automata and Computability

Quiz 3

Section 16 (Set A)

Full name:

ID:

Total Marks: 20

1. Prove that the following languages are non-regular using pumping lemma.
 - a. $\{a^n b^l c^k \mid k \neq n + l\}$
 - b. $\{0^n 1^m : n = 2q + r_1, m = 2q + r_2, q \geq 0, 0 \leq r_1, r_2 \leq 3\}$

2. Write the CFG of the following languages, then show derivation of a string of length at least 5 from the CFG:
- $\Sigma = \{a, b\}$, number of a is a multiple of 4.
 - $\{a^i b^j c^k \mid i > 2j + 3k\}$

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Quiz 4

Section 16 (Set A)

Full name:

ID:

Total Marks: 20

1. Write CFG of the following:

- a. $\Sigma = \{a, b\}$, contains an even palindrome as a substring.
- b. $L = \{w_1 \# w_2 \mid |w_1| \text{ is odd, } w_2 \text{ contains equal number of 0s and 1s}\}$, $\Sigma = \{0, 1\}$

2. Answer the questions from the given CFG:

$$S \rightarrow abA \mid aB \mid abS$$
$$A \rightarrow cS \mid c$$
$$B \rightarrow bcS \mid b$$

- Write a leftmost derivation and show the parse tree of the string `abcbcab`
- Write two other leftmost derivations and parse trees of the given string (except for the one you wrote in (a)) to show that the grammar is ambiguous.
- Write one string of length at least 6 which only has one derivation, also show the derivations.

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Quiz 3

Section 16 (Set B)

Full name:

ID:

Total Marks: 20

1. Prove that the following languages are non-regular using pumping lemma.
 - a. $\{a^n b^m \mid n \neq m\}$
 - b. $\{0^n 1^m : n = 4q + r_1, m = 4q + r_2, q \geq 0, 0 \leq (r_1, r_2) \leq 5\}$

2. Write the CFG of the following languages, then show derivation of a string of length at least 5 from the CFG:
- $\Sigma = \{a, b\}$, number of b is a multiple of 3.
 - $\{a^i b^j c^k \mid i > 3j + 2k\}$

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Quiz 4

Section 16 (Set B)

Full name:

ID:

Total Marks: 20

1. Write CFG of the following:

a. $\Sigma = \{a, b\}$, contains an odd palindrome as a substring.

b. $L = \{w_1 \# w_2 \mid |w_2| \text{ is even, } w_1 \text{ contains equal number of 0s and 1s}\}$, $\Sigma = \{0, 1\}$

2. Answer the questions from the given CFG:

$$S \rightarrow deA \mid dB \mid deS$$
$$A \rightarrow fS \mid f$$
$$B \rightarrow efS \mid e$$

- Write a leftmost derivation and show the parse tree of the string defdefde.
- Write two other leftmost derivations and parse trees of the given string (except for the one you wrote in (a)) to show that the grammar is ambiguous.
- Write one string of length at least 6 which only has one derivation, also show the derivations.